Jatin Pathak

Mumbai, Maharashtra

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SUMMARY:

Electronics & Telecommunication engineering student with hands-on experience in backend, full-stack, AI/GenAI, and cloud development. Skilled in Node.js, Express.js, React, and MongoDB, with practical exposure to LLMs, RAG, and AWS. Interested in building secure, scalable applications and applying software, AI, and cloud technologies to real-world problems.

EDUCATION:

THAKUR COLLEGE OF ENGINEERING AND TECHNOLOGY ***Mumbai, India***

B.E. in Electronics & Telecommunication (CGPA – 7.92) *2023-2027*

COURSEWORK / SKILLS:

- **Language:** C++, TypeScript, JavaScript, Python (Basics)
- **AI / GenAI:** LLMs, RAG, LangChain, Prompt Engineering, Tool Calling
- **Web Dev:** HTML, CSS, Tailwind CSS, React(Intermediate), Node.js, Express.js
- **Database:** PostgreSQL, MongoDB, MySQL(Basics)
- **Cloud / DevOps:** AWS (EC2, S3, IAM), Docker, GitHub Actions
- **Dev Tools:** Cursor, Antigravity, Git, GitHub, Postman, VS Code
- **Core Concepts:** REST APIs, DBMS, CN

EXPERIENCE:

Backend Developer Intern | *Bluestock Fintech* ***2 Months***

- Developed backend services using **Node.js**, **Express.js**, and **MongoDB** for contact forms, early-access registrations, and dashboard data.
- Built **5+ REST APIs** for contact submissions, subscriptions, and dashboard functionality.
- Implemented **input validation**, error handling, and duplicate email checks for reliable API operations.
- Tested and debugged APIs using **Postman** and managed code with **Git/GitHub**.

PROJECTS:

Deep Packet Inspection | Node.js, TypeScript, Wireshark, Worker Threads (GitHub)

- Built a stateful DPI firewall to inspect TLS SNI and network protocols for application identification and filtering..
- Implemented multi-threaded packet processing with 5-tuple flow tracking, TCP session management, and Vitest-based validation.

Vault Backend | Node.js, Express.js, MongoDB, JWT, Nodemailer (GitHub)

- Built a secure money transfer backend with RESTful APIs, a 10-step transaction workflow, balance validation, transaction tracking, and automatic rollback.
- Secured APIs using JWT authentication, MongoDB transactions, and Nodemailer for email notifications.

CipherShield AI | React, Node.js, Express, MongoDB, REST APIs (GitHub | Live)

- Built a cybersecurity platform for secure text, file, and image encryption, featuring JWT authentication and an admin dashboard.
- Developed a cryptographic engine using NPCR and UACI metrics to evaluate and benchmark encryption security.

EXTRACURRICULAR:

TCET-IEEE | Working Committee

- Planned technical events, prepared technical content, and coordinated with the team for successful execution.

ACADEMIC TUTOR

- Mentored 25+ students as an academic tutor, enhancing communication and interpersonal abilities

CERTIFICATION:

- **JavaScript Bootcamp** — LetsUpgrade LUEJSSEP124466